# Bank Loan Default Analysis

## Default Prediction & Credit-Risk Insights

Business Case Study • Dataset: **Default_Fin.csv** (10,000 records)
Prepared for: Credit Risk & Lending Decisions

| 10,000 | 3.33% | 333 | 70.56% |
|---|---|---|---|
| Total customers | Default rate | Defaulters | Employed |

*Figures describe the supplied dataset and are associations, not causal relationships or statements about the bank's live portfolio.*

# 1. Executive Summary

This report analyses **10,000 customer records** from **Default_Fin.csv** to understand which financial characteristics are associated with loan default. The observed default rate is **3.33%** (333 defaulters) — a highly imbalanced target, roughly 1 defaulter for every 29 non-defaulters.

The single most important finding is that **bank balance is the dominant signal for default in this dataset**. Default risk is near-zero below 5,000 but rises to **40.85%** above 20,000. Counter-intuitively, defaulters hold about **2.2× the average balance** of non-defaulters (20,974 vs 9,647). Employment matters far less, and salary shows almost no monotonic relationship.

### Key takeaways

- **Overall default rate 3.33%** — 333 of 10,000 customers.
- **Unemployed default more**: 4.31% vs 2.92% employed — real but modest.
- **Bank balance is decisive**: 0% (<5k) → 40.85% (>20k); correlation 0.35.
- **Salary is weak**: near the 3.33% baseline across all bands (correlation -0.02).
- A baseline model reaches **ROC-AUC 0.948**, recall 0.83 — features rank risk well, chiefly via balance.

### Bottom line

Bank balance alone separates most future defaulters, so it should anchor any early-warning rule. But the dataset lacks credit scores, loan amounts and repayment history, so these findings should **inform** — not automate — lending decisions, and must be checked for fair-lending compliance before operational use.

# 2. Business Background & Objectives

A bank lends to customers with varying employment statuses, bank balances and annual salaries. Some borrowers fail to repay, causing losses. The bank wants to analyse historical data to understand factors associated with default, identify high-risk customers, and make more informed lending decisions.

### Specific objectives

- Measure the overall loan default rate.
- Compare default rates between employed and unemployed customers.
- Investigate how bank balance and annual salary relate to default risk.
- Identify segments needing extra credit-risk assessment.
- Develop a data-driven approach to support loan-approval decisions.
- Lay a foundation for a future probability-of-default model.

### Business questions addressed

| # | Business question | Where answered |
|---|---|---|
| Q1 | What is the overall default rate? | §3, §4 |
| Q2 | Does employment status relate to default risk? | §5 |
| Q3 | How does bank balance relate to default? | §6 |
| Q4 | Does annual salary relate to default risk? | §7 |
| Q5 | Which characteristics distinguish defaulters? | §6–§8 |
| Q6 | Can the bank identify high-risk customers? | §9 (model) |

# 3. Dataset Overview & Data Cleaning

## Deliverable 1 — Data Cleaning & Exploratory Data Analysis

The dataset contains 10,000 records and 5 columns with no missing values. Each row is one customer.

| Column | Type | Description |
|---|---|---|
| Index | Integer | Sequential record identifier |
| Employed | Binary | 1 = employed, 0 = unemployed |
| Bank Balance | Numeric | Recorded bank balance (currency unspecified) |
| Annual Salary | Numeric | Annual salary (currency unspecified) |
| Defaulted? | Binary | Target: 1 = default, 0 = no recorded default |

### Data-quality validation

| Check | Result |
|---|---|
| Total records | 10,000 |
| Total columns | 5 |
| Missing values | 0 (all columns) |
| Duplicate rows (full / excl. Index) | 0 / 0 |
| Recorded defaulters | 333 |
| Non-defaulters | 9,667 |
| Binary fields valid (0/1 only) | Employed ✓, Defaulted? ✓ |
| Negative values | None |
| Target | Defaulted? (imbalanced, 3.33%) |

### Descriptive statistics

| Metric | Bank Balance | Annual Salary |
|---|---|---|
| Mean | 10,024 | 402,204 |
| Median | 9,884 | 414,632 |
| Minimum | 0 | 9,264 |
| Maximum | 31,852 | 882,651 |

*A number of customers record a bank balance of exactly 0, treated as valid; salary has a floor near 9,264.*

## 4. Overall Default Rate (Q1)

Of 10,000 customers, **333 defaulted** (3.33%) and **9,667 did not** (96.67%). This heavy imbalance is the central challenge: a model predicting 'no default' for everyone would be 96.67% accurate yet useless — so accuracy must be replaced by precision, recall, F1 and PR-AUC when evaluating any model.

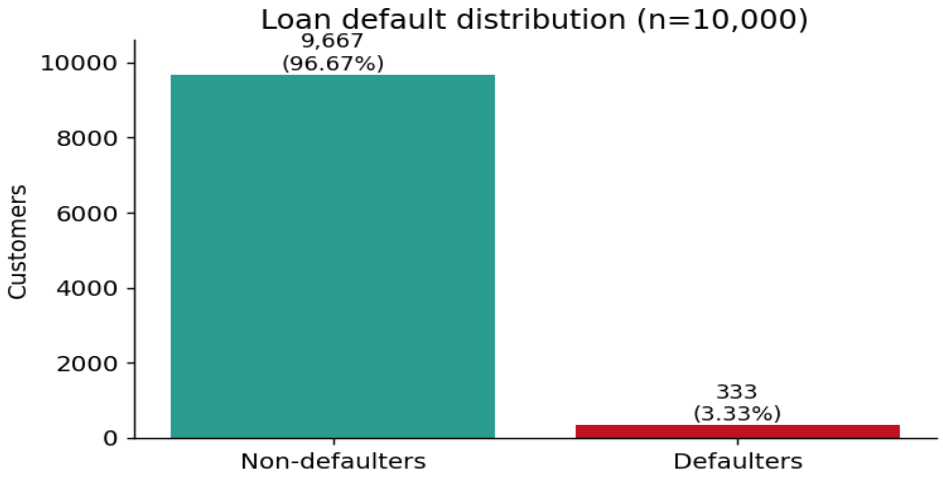



Figure 1 — Default vs non-default distribution across the 10,000 customers.

## 5. Default Rate by Employment Status (Q2)

Unemployed customers default at **4.31%** versus **2.92%** for employed — roughly 1.5× more likely. A genuine but modest difference; employment alone is a weak classifier because both rates are low.

| Employment | Customers | Defaulters | Default rate |
|---|---|---|---|
| Unemployed (0) | 2,944 | 127 | 4.31% |
| Employed (1) | 7,056 | 206 | 2.92% |
| **All** | **10,000** | **333** | **3.33%** |

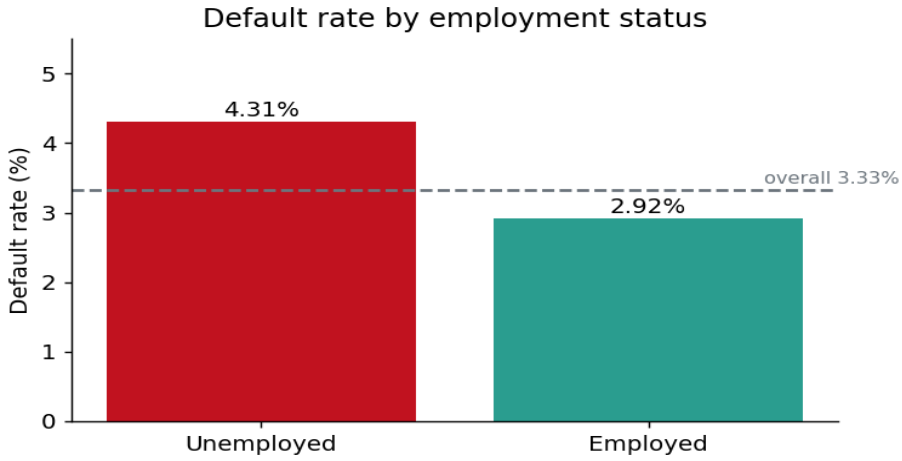



Figure 2 — Default rate by employment status vs the 3.33% baseline.

# 6. Bank Balance & Default Risk (Q3, Q5)

Bank balance is the strongest signal in the dataset. Default rate escalates steeply and monotonically with balance — from **0% below 5,000** to **40.85% above 20,000**. Correlation with default is **0.35**, by far the largest of any feature.

| Balance band | Customers | Defaulters | Default rate |
|---|---|---|---|
| 0–5k | 2,118 | 0 | 0.0% |
| 5k–10k | 2,962 | 3 | 0.1% |
| 10k–15k | 2,921 | 23 | 0.79% |
| 15k–20k | 1,502 | 104 | 6.92% |
| 20k+ | 497 | 203 | 40.85% |

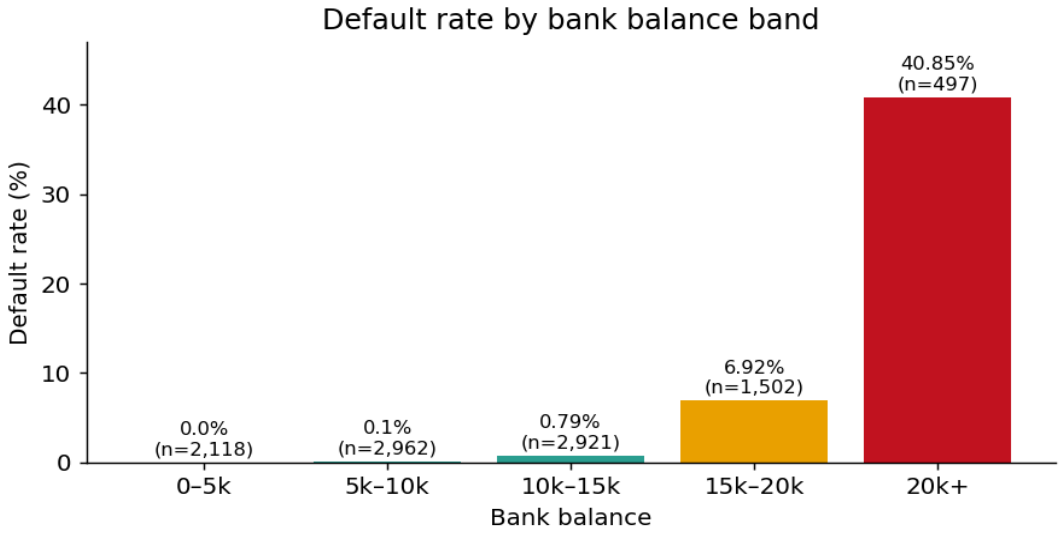



Figure 3 — Default rate rises sharply with bank balance; the 20k+ band concentrates most defaults.

Defaulters hold a far higher average balance (20,974) than non-defaulters (9,647); the distributions below show defaulters shifted strongly to the right.

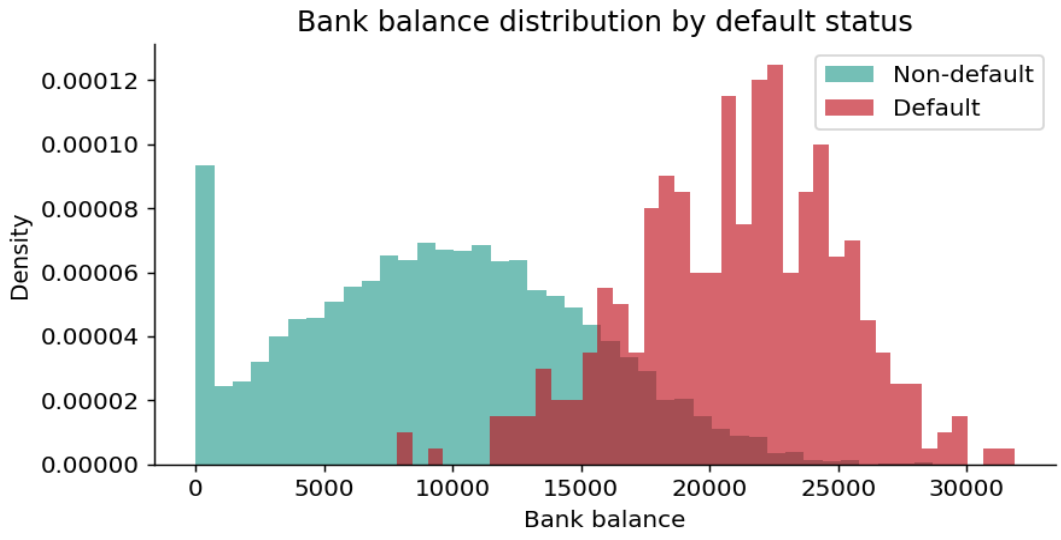



Figure 4 — Bank balance distribution: defaulters (red) cluster at much higher balances.

### Interpretation & caution

This striking association must be read carefully. A high recorded balance coinciding with default may reflect how this dataset was constructed, or an unobserved factor. It does **not** prove that holding more money causes default — but it does make balance an effective *ranking* variable for flagging accounts for review.

# 7. Annual Salary & Default Risk (Q4)

Annual salary shows almost no relationship with default. Across all bands the rate stays near the 3.33% baseline, and correlation with default is only **-0.02** (effectively zero). Salary is not a useful standalone predictor.

| Salary band | Customers | Defaulters | Default rate |
|---|---|---|---|
| <150k | 363 | 13 | 3.58% |
| 150k–300k | 2,881 | 119 | 4.13% |
| 300k–450k | 2,546 | 73 | 2.87% |
| 450k–600k | 3,074 | 87 | 2.83% |
| 600k+ | 1,136 | 41 | 3.61% |

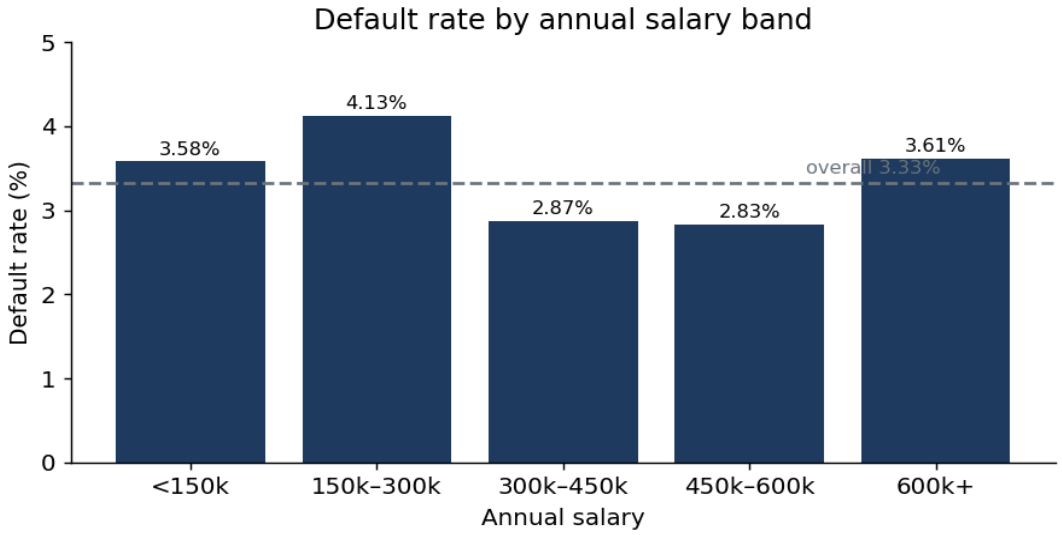



Figure 5 — Default rate is essentially flat across salary bands.

# 8. What Distinguishes Defaulters (Q5)

| Metric | Non-defaulters | Defaulters |
|---|---|---|
| Avg bank balance | 9,647 | 20,974 |
| Median bank balance | 9,634 | 21,469 |
| Avg annual salary | 402,794 | 385,070 |
| Median annual salary | 415,074 | 378,184 |

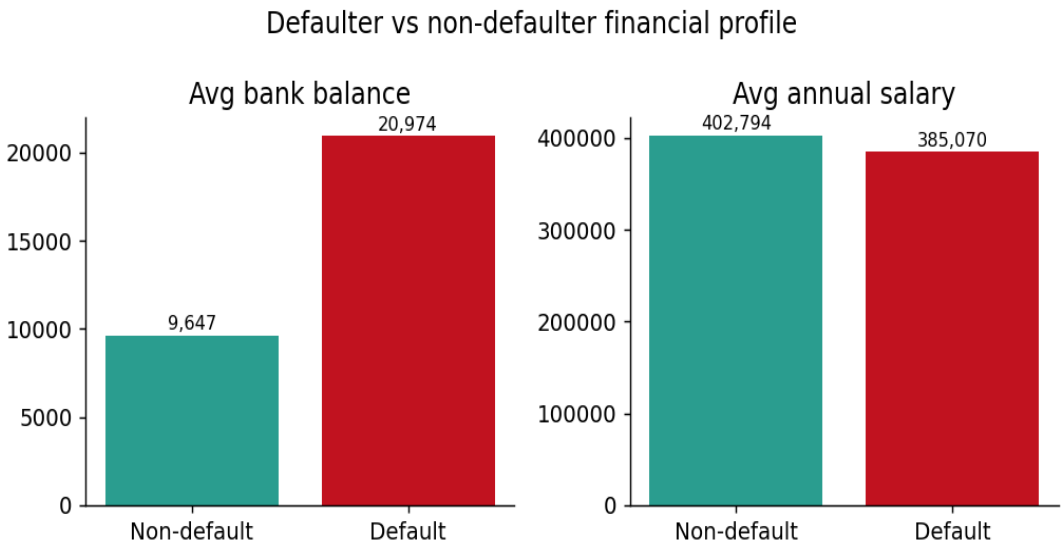



Figure 6 — Defaulters hold ~2.2× the average balance but similar salary.

The defining difference is **bank balance**, not salary. A simple at-risk profile is **high bank balance, unemployed or lower-salaried** — with balance doing most of the discriminating work.

# 9. Baseline Default-Prediction Model (Q6)

## Deliverable 4 — Optional Default Prediction Model

A baseline **logistic regression** was trained on a 70/30 stratified split with **class weighting** to correct the 3.33% imbalance and standardised features. Given the imbalance, it is judged on precision, recall, F1 and PR-AUC rather than accuracy.

| Metric | Value | Reading |
|---|---|---|
| ROC-AUC | 0.948 | Excellent ranking of risk |
| PR-AUC | 0.549 | Strong vs 0.033 baseline |
| Recall (defaulters caught) | 0.83 | Catches most true defaulters |
| Precision | 0.185 | Many false alarms at 0.5 cutoff |
| F1-score | 0.303 | Precision/recall trade-off |

On the held-out test set the model correctly flagged **83 of 100** actual defaulters (recall 0.83), at the cost of 365 false alarms. The high ROC-AUC (0.948) confirms the features — led overwhelmingly by bank balance — rank customers by risk very effectively. Low precision at the 0.5 threshold is expected under imbalance and is a **threshold-tuning** decision, not a model failure.

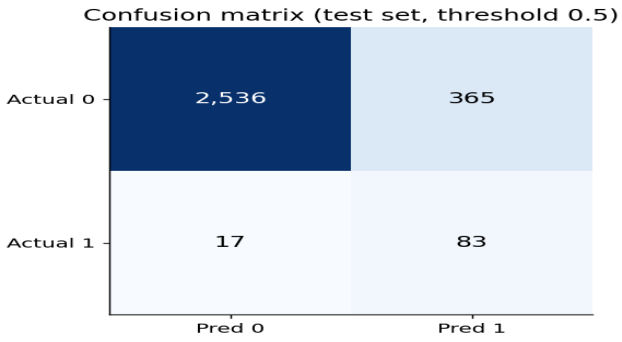



Figure 7 — Confusion matrix on the test set (threshold 0.5).

**Feature contribution** (standardised coefficients): **Bank Balance 2.843** (dominant), Employed 0.262, Annual Salary 0.036 (negligible). The bank can identify high-risk customers from these features, but the predictive power is almost entirely a balance effect — this validates signal, it is not a production credit model.

# 10. Interactive Business Dashboard

## Deliverable 2 — Dashboard specification

The following can be built directly in Power BI or Tableau from Default_Fin.csv, presenting the KPIs and segment comparisons as an interactive view.

### KPI cards

- Total Customers — 10,000
- Total Defaulters — 333
- Default Rate — 3.33%
- Employment Rate — 70.56%
- Avg Bank Balance — 10,024
- Avg Annual Salary — 402,204

### Visuals

- Bar — default rate by employment status.
- Column — default rate by bank-balance band.
- Column — default rate by salary band.
- Histogram — balance distribution split by default status.
- Donut — defaulters vs non-defaulters share.
- Table — segment drill-down (employment × balance band).

### Slicers / filters

- Employment status
- Bank-balance band
- Salary band
- Default flag

### KPI definitions (data model)

| KPI | Definition |
|---|---|
| Default Rate (%) | Defaulters ÷ Total Customers × 100 |
| Non-Default Rate (%) | Non-defaulters ÷ Total Customers × 100 |
| Employment Rate (%) | Employed ÷ Total Customers × 100 |
| Default Rate by Employment | Default rate within each employment group |
| Default Rate by Balance Band | Default rate within each balance band |
| Default Rate by Salary Band | Default rate within each salary band |

# 11. Constraints, Assumptions & Limitations

| Area | Constraint / assumption |
|---|---|
| Dataset scope | Limited to the 10,000 supplied records. |
| Target variable | Defaulted?=1 is a recorded default; exact definition not provided. |
| Employment | Employed=1 means employed; 0 means unemployed. |
| Missing data | No missing values found. |
| Data quality | No duplicates or negative values; balances of 0 treated as valid. |
| Currency | Currency of salary and balance is unspecified. |
| Causation | Findings are associations, not causal relationships. |
| Predictive modelling | Any model needs validation, imbalance handling and proper metrics. |
| Business costs | Cost of approving a defaulter vs rejecting a good customer is unknown. |
| Fair lending | Features must be checked for fairness and regulatory compliance before use. |

### Key analytical challenges

- **Class imbalance** — only 3.33% defaulters, so accuracy is misleading; use recall / PR-AUC.
- **Limited features** — no credit score, loan amount, repayment history, tenure or debt-to-income ratio.
- **Correlation ≠ causation** — especially the counter-intuitive balance effect.
- **Threshold choice** — weigh missed defaults against false alarms on good customers.
- **No loss quantification** — without loan amounts or dates, recovery rates, overdue days and total losses cannot be computed.

# 12. Recommendations & Next Steps

## Deliverable 5 — Executive summary & recommendations

### Most important findings

- Portfolio default rate is **3.33%**; defaults are concentrated, not spread evenly.
- **Bank balance is the dominant risk marker** — the 20k+ band has a 40.85% default rate and holds most defaulters.
- **Unemployment adds moderate risk** (4.31% vs 2.92%).
- **Salary is not predictive** on its own.

### Recommended actions

- **Introduce a balance-based early-warning flag.** Route high-balance accounts (e.g. 15k+) to enhanced review rather than auto-decisioning.
- **Use the model for ranking, not rejection.** Score and rank applicants by default probability to prioritise manual review.
- **Tune the decision threshold to business cost.** Set the cutoff to minimise expected cost once loss/profit figures are known, not 0.5.
- **Investigate the balance–default relationship** before operationalising it.
- **Monitor for fair-lending risk** — validate the features create no prohibited disparate impact.

### Additional data needed before lending decisions

- Credit score / bureau history
- Loan amount, tenure and product type
- Repayment history and days-past-due
- Debt-to-income ratio and existing obligations
- Dates (origination, default) for recovery and loss
- Cost parameters (loss given default, profit on a good loan)

### Final outcome

The bank now has a clear, quantified picture of the observed default rate, the strong role of bank balance, the modest role of employment and the weak role of salary — with an explicit statement of the dataset's limits. These findings should **support better-informed, human-reviewed risk assessment** and provide a validated foundation for a future, richer predictive model once additional credit data is available — not, on their own, automate loan approvals or rejections.